GYR-O Foundations

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Summary

GYR-O's reduction of visual clutter rests on one well-established fact: human attention and working memory are limited. Every item on a page competes for a glance and for a slot in short-term memory, so what the reader does not need still costs them something.

Eye-movement research is how that cost is measured. A cluttered page produces more fixations, longer searches and more backward jumps (regressions). GYR-O's principles map onto seven bodies of research, listed below. They support the principles; they do not yet prove that GYR-O itself performs better.

Principles and their foundations

Each GYR-O principle, the research it rests on, and the cost it avoids.

GYR-O principle	Rests on	Cost avoided
First-time visitor is the highest priority	Information foraging	Leaving before finding the answer
Intro is self-contained; a brief summary is often enough	Information foraging; cognitive load theory	Reading further than needed
Intro on/off, in the visitor's hands ("ergonomics")	Cognitive load theory; visual search	Re-scanning material already known
No "hunting"	Visual search and crowding; eye movements in reading	Extra fixations and search time
Information comes to the reader	Information foraging ("information scent")	Navigation and guessing where to look
Scroll-to-text	Eye movements in reading; visual search	Locating the passage by eye
Fewer elements per screen	Hick-Hyman law; Tufte's data-ink ratio	Decision time and ink that carries nothing

GYR-O principle	Rests on	Cost avoided
The owner never has to know what is under the hood	Information hiding (Parnas)	Learning hosting, domains and code just to say something

The Intro/Info switch

Two checkboxes give four states, so one page serves both the ultra-compact reader and the detailed reader without either getting in the other's way. The page works like a header and a body with no footer: Intro is the header, Info is the body.

The four states, each named for a familiar kind of document.

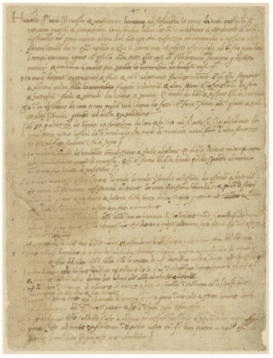
Intro	Info	What shows	Named for
off	off	Thumbnail of the signature graphic only	Card
on	off	Intro and the signature graphic	Handbill
off	on	Thumbnail of the signature graphic and Info	Reference
on	on	Intro, signature graphic and Info	Full Form

"Intro" is the everyday word; what it is, in fact, is the **synopsis**. Often the synopsis is all that is necessary or desired, and anything more is visual clutter. GYR-O calls the formats built on this idea **synoptic formats**, from the Greek for "seeing all together": the whole taken in at one view. A synoptic format is concise and self-contained. A business card, a one-page resume, a circus poster, a handbill, a broadsheet, a highway billboard and a map are all synoptic formats. Whichever one a site's Intro takes is the whole Intro, complete in itself.

Synoptic formats: concise and self-contained, taken in at one view



Business card
Trade card, 19th c. · Met · CC0



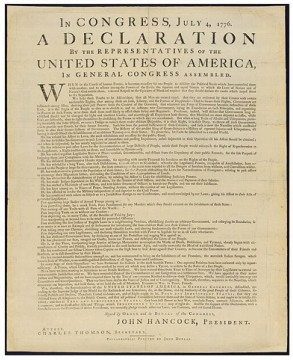
One-page resume
Leonardo to Sforza, c.1482 · PD



Circus poster
Barnum & Bailey, 1899 · PD



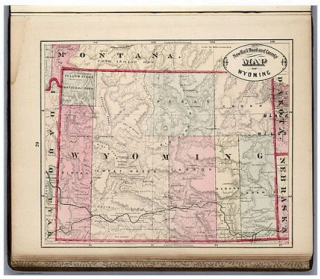
Handbill
Mammoth Tree, 1854 · PD



Broadsheet
Dunlap broadside, 1776 · PD



Highway billboard
North Dakota · Chitrapa · CC BY-SA 4.0



Map
Cram, rail and county map of Wyoming · PD

Seven synoptic formats, from Wikimedia Commons; click a name to see it full size: trade card (Met, CC0); Leonardo to Ludovico Sforza (public domain); Barnum & Bailey poster (public domain); Mammoth Tree handbill (public domain); Dunlap broadside (Library of Congress, public domain); highway billboard (Chitrapa, CC BY-SA 4.0); rail and county map of Wyoming (George F. Cram, public domain).

A resume-format Intro shows how short the compact read can be. In Ladders' 2018 eye-tracking study, recruiters' initial screen of a resume averaged 7.4 seconds, up from about 6 seconds in its 2012 study. They looked first for name, current and previous titles, and dates. Resumes that did poorly had long sentences, multiple columns and little white space (Ladders 2018 PDF; HR Dive). Caveat: this was an industry study, not peer-reviewed, and the 2018 report does not give its sample size (ERE).

The design term for this is **progressive disclosure** (Nielsen): show the essentials first and let the reader open the detail. GYR-O's version is unusual because the reader can also close the essentials once they are known. That is the cognitive-load argument in the research section, put in the visitor's hands.

Rule: the signature graphic is visible in every state, at full size or as a thumbnail, and is never left out. The reason is dual coding, one of GYR-O's foundations: words and pictures are remembered by two linked routes (Paivio), so a page that always shows both is remembered better and serves the visual reader as well as the verbal one.

Precedent: the Holy Grail layout

GYR-O's three columns have a well-known ancestor. The "Holy Grail" is the classic web page: a header, a footer, and three columns between them, with navigation on the left, the main content in the center and extras such as ads on the right. Matthew Levine's 2006 article in *A List Apart* is its standard write-up, and for years it was hard to build; CSS Grid has since made it easy (Levine 2006).

GYR-O keeps the shape and changes what it means:

- **No footer.** The page is a header (the Intro) and a body (the Info).
- **The sources take the center.** The Holy Grail centers the main content and pushes support to the sides; GYR-O centers the sources, at full weight, between the plain answers and the technical detail.
- **Every column carries content.** There is no navigation or ad column; each column serves a different reader, and each can be switched off.
- **The header can be switched off.** The Holy Grail's header is fixed; GYR-O's Intro, the synopsis, is the reader's choice, giving the four states from Card to Full Form.

Both stack into a single column on a phone.

The research

Eye movements in reading. Reading is not a smooth sweep. The eye jumps (saccades, about 7 to 9 letters forward), pauses to take in text (fixations, roughly 200 to 250 ms), and sometimes jumps back to reread (regressions). Javal observed the jumps in 1879; Huey described them in 1908; Keith Rayner's work is the modern standard. These are the units in which a page's cost to the reader can be counted.

Visual search and crowding. The time to find a target grows with the number of other items on screen, and items packed close together interfere with each other (Treisman; Wolfe). This is the science behind "hunting."

Cognitive load theory. Working memory holds only a few items at once. Material the reader does not need uses capacity that understanding needs (Sweller). Letting the visitor switch Intro off removes load they have already paid.

Information foraging. Readers follow "information scent" toward what they want and give up when searching costs more than it returns (Pirolli and Card). GYR-O's "information comes to the reader" is this principle stated as design: shorten the path so the scent never has to be followed far.

Hick-Hyman law. Decision time rises with the number of choices offered. Fewer options on screen means a faster first move.

Tufte's data-ink ratio. Remove whatever on the page carries no information. Tufte's term for decoration that does no work is "chartjunk."

Information hiding. Each part of a system hides its inner workings behind a simple surface, so whoever uses it never needs to know them (Parnas). This principle serves the site's owner rather than its reader: Auto-Gyro handles domain names, hosting, certificates, code and publishing out of sight, and the owner says what the site is for and decides what it says.

Grounded, not yet proven

The research shows that clutter and hunting cost the reader. It does not show that a GYR-O page costs less than a conventional one. Until that is measured, GYR-O should say its principles are *grounded in* this research, not *proven by* it.

A simple test would settle it:

1. Pick one site's content and build it two ways: a GYR-O page and a conventional page.
2. Give first-time readers the same three questions to answer from the page.
3. Time each answer and count wrong answers and give-ups.
4. If an eye tracker is available (webcam-based ones are inexpensive), also count fixations and regressions per answer.
5. Repeat with returning readers, Intro switched off, to test the on/off principle separately.

Sources

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